

A Project Report
On
**AI-Powered Intrusion Detection
System**

BY
Gouri Mandhani and Shreya Talod

Se22uari215 and Se22uari171

Under the supervision of

Raghu Kishore Sir

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Certificate

This is to certify that the project report entitled “**TITLE**” submitted by **Mr/Ms. STUDENT NAME** (ID No. **xxxxxxxxxx**) in partial fulfillment of the requirements of the course PR4101, Project Course, embodies the work done by him/her under my supervision and guidance.

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Date:

ABSTRACT

With the rapid increase in cyber threats and malicious network activities, intelligent Intrusion Detection Systems (IDS) have become essential for ensuring network security. Traditional rule-based IDS approaches often struggle to detect evolving and complex cyberattacks effectively. This project presents an AI-Powered Intrusion Detection System using machine learning, deep learning, transformer architectures, and Explainable AI techniques for multi-class cyberattack detection.

The proposed system was developed and evaluated using two benchmark cybersecurity datasets: UNSW-NB15 and CIC-IDS Collection. Several preprocessing techniques including data cleaning, label encoding, feature scaling, and dataset balancing were applied before model training. Multiple artificial intelligence models such as Decision Tree, Random Forest, XGBoost, Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Transformer-based architectures were trained and compared for intrusion detection performance.

The models were evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, and confusion matrix analysis. Experimental results demonstrated that ensemble learning approaches, particularly Random Forest and XGBoost, achieved superior overall performance compared to deep learning and transformer-based models on structured network traffic data. Deep learning models showed strong sequential learning capabilities, while the Transformer model demonstrated comparatively lower performance on tabular intrusion detection datasets.

In addition, SHAP-based Explainable AI was integrated to improve model interpretability and identify the most influential network traffic features contributing to cyberattack classification decisions. The project highlights the effectiveness of AI-driven intrusion detection systems and demonstrates the importance of explainable and interpretable cybersecurity solutions for modern network environments.

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CHAPTER - 1 - INTRODUCTION

The rapid growth of internet-based services, cloud computing, IoT devices, and digital communication has significantly increased the risk of cyber threats and malicious network activities. Cyberattacks such as Denial of Service (DoS), Distributed Denial of Service (DDoS), Botnet attacks, infiltration attacks, and web-based attacks can compromise sensitive information, disrupt network services, and cause major financial and organizational losses. As a result, ensuring network security has become one of the most important challenges in modern cybersecurity systems.

Intrusion Detection Systems (IDS) are designed to monitor network traffic and detect suspicious or malicious activities within computer networks. Traditional rule-based IDS approaches rely heavily on predefined attack signatures and manually created rules. Although such systems are effective for detecting known attacks, they often struggle to identify new, evolving, and sophisticated cyber threats. In addition, traditional systems may generate high false alarm rates and lack adaptability in dynamic network environments.

Artificial Intelligence (AI) and Machine Learning (ML) techniques have emerged as powerful solutions for improving intrusion detection capabilities. AI-based IDS models can automatically learn complex attack patterns from network traffic data and improve detection accuracy for both known and unknown attacks. Deep learning and transformer-based architectures further enhance the ability to capture sequential and contextual relationships in network traffic behavior.

This project presents an AI-Powered Intrusion Detection System using machine learning, deep learning, transformer architectures, and Explainable AI techniques for multi-class cyberattack detection. The proposed system was evaluated using the UNSW-NB15 and CIC-IDS Collection datasets, which contain both normal and malicious network traffic records across multiple attack categories.

Several artificial intelligence models including Decision Tree, Random Forest, XGBoost, Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Transformer-based architectures were trained and compared using evaluation metrics such as Accuracy, Precision, Recall, F1-score, ROC-AUC, and confusion matrix analysis. In addition, SHAP-based Explainable AI was used to improve model interpretability and identify the most influential network traffic features affecting intrusion detection decisions.

The main objective of this project is to develop an intelligent and interpretable intrusion detection system capable of accurately identifying multiple cyberattacks while comparing the effectiveness of different AI architectures on benchmark cybersecurity datasets.

CHAPTER - 2 -- PROBLEM DEFINITION

The rapid growth of digital communication, cloud computing, IoT devices, and internet-based services has significantly increased the number of cyber threats targeting modern computer networks. Cyberattacks such as Denial of Service (DoS), Distributed Denial of Service (DDoS), Botnet attacks, infiltration attacks, and web-based attacks can compromise sensitive information, disrupt services, and cause severe financial and organizational losses. As a result, protecting network infrastructure from malicious activities has become a major challenge in cybersecurity.

Traditional Intrusion Detection Systems (IDS) mainly rely on predefined signatures and manually designed rules to detect cyberattacks. Although such systems can identify known attacks, they often fail to detect unknown or evolving threats effectively. In addition, traditional IDS approaches may generate high false positive rates and lack adaptability in dynamic network environments. These limitations create the need for intelligent intrusion detection systems capable of automatically learning complex attack patterns from network traffic data.

The main objective of this project is to develop an AI-Powered Intrusion Detection System capable of detecting and classifying multiple cyberattacks using machine learning, deep learning, transformer architectures, and Explainable AI techniques. The proposed system aims to improve intrusion detection accuracy, reduce false classifications, and provide interpretable predictions for cybersecurity analysis.

The project focuses on answering the following research questions:

- Which artificial intelligence model performs best for multi-class intrusion detection?

- Can ensemble learning approaches outperform deep learning and transformer-based architectures on structured network traffic datasets?
- How effectively can AI models classify different cyberattack categories such as DoS, DDoS, Botnet, and Web attacks?
- Can Explainable AI improve the interpretability and transparency of intrusion detection systems?

The proposed intrusion detection system performs multi-class classification on network traffic data obtained from the UNSW-NB15 and CIC-IDS Collection datasets. The input network traffic feature vector can be represented as:

$$\mathbf{X} = \{x_1, x_2, x_3, \dots, x_n\}$$

where each x_i represents a network traffic feature such as packet size, flow duration, packet rate, packet timing, or traffic statistics.

The intrusion detection model learns a mapping function:

$$\mathbf{f}(\mathbf{X}) \rightarrow \mathbf{Y}$$

where $\mathbf{Y}$ represents the predicted attack class label:

$$\mathbf{Y} \in \{\text{Benign, DoS, DDoS, Botnet, Web Attack, Infiltration, Brute Force, PortScan}\}$$

Several challenges are associated with intrusion detection in modern cybersecurity environments. These include high-dimensional network traffic data, class imbalance between attack categories, continuously evolving attack patterns, false positive classifications, and the lack of interpretability in complex AI models. Therefore, developing an accurate and interpretable intrusion detection framework remains an important research problem.

To address these challenges, this project compares multiple machine learning, deep learning, and transformer-based architectures including Decision Tree, Random Forest, XGBoost, RNN, LSTM, and Transformer models. The models are evaluated using Accuracy, Precision, Recall, F1-score,

ROC-AUC, and confusion matrix analysis. In addition, SHAP-based Explainable AI is integrated to identify important network traffic features and improve the transparency and interpretability of intrusion detection decisions.

The proposed system aims to provide an efficient, scalable, and intelligent cybersecurity solution capable of accurately identifying multiple cyberattacks while improving trust and interpretability in AI-driven intrusion detection systems.

CHAPTER - 3 -- BACKGROUND AND RELATED WORK

3.1 Overview of Intrusion Detection Systems

Intrusion Detection Systems (IDS) are used to monitor network traffic and detect malicious activities within computer networks. Traditional IDS approaches mainly rely on signature-based techniques that identify attacks using predefined rules. Although effective for known attacks, these systems often fail to detect new and evolving cyber threats.

To overcome these limitations, researchers have increasingly applied Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) techniques for intrusion detection. These approaches automatically learn attack patterns from network traffic data and improve attack classification performance.

3.2 Related Work on Machine Learning-Based IDS

Several studies have applied machine learning algorithms such as Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost for intrusion detection systems. Random Forest and XGBoost achieved high classification accuracy because ensemble learning methods reduce overfitting and improve robustness.

Researchers using the UNSW-NB15 and CIC-IDS datasets reported that ensemble learning approaches perform effectively for multi-class cyberattack detection. However, many existing models suffer from class imbalance issues and limited interpretability.

Naive Bayes and SVM-based IDS models demonstrated low computational complexity, but their performance decreased on highly complex and large-scale network datasets.

3.3 Related Work on Deep Learning-Based IDS

Deep learning models such as Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks have been widely used for intrusion detection because they can learn sequential patterns from network traffic data.

LSTM models improve long-term dependency learning and overcome the vanishing gradient problem associated with traditional RNNs. Several studies reported strong intrusion detection performance using LSTM architectures on benchmark cybersecurity datasets.

However, deep learning models require higher computational resources and longer training time. In addition, many deep learning models behave as black-box systems with limited interpretability.

3.4 Transformer and Explainable AI Approaches

Transformer architectures use attention mechanisms to learn contextual relationships between input features. Recent research explored the use of Transformer models for intrusion detection and demonstrated promising results in sequential learning tasks.

However, Transformer models are computationally expensive and may not always outperform ensemble learning methods on structured tabular network traffic data.

To improve model transparency, researchers have applied Explainable Artificial Intelligence (XAI) techniques such as SHAP (SHapley Additive exPlanations). SHAP helps identify the most important network traffic features influencing intrusion detection decisions and improves model interpretability.

3.5 Research Gaps and Proposed Work

Although many intrusion detection models achieve high accuracy, several challenges still remain. Existing systems may suffer from poor generalization, class imbalance problems, high computational complexity, and lack of interpretability.

To address these limitations, this project compares machine learning, deep learning, and transformer-based architectures including Decision Tree, Random Forest, XGBoost, RNN, LSTM, and Transformer models using the UNSW-NB15 and CIC-IDS datasets. In addition, SHAP-based Explainable AI is integrated to improve transparency and analyze feature importance.

3.6 Mathematical Overview of Models

The proposed intrusion detection system performs multi-class classification on network traffic features represented as:

$$\mathbf{X} = \{x_1, x_2, x_3, \dots, x_n\}$$

where x_i represents network traffic features such as packet size, flow duration, and traffic statistics.

The classification model learns a mapping function:

$$f(\mathbf{X}) \rightarrow \mathbf{Y}$$

where $\mathbf{Y}$ represents the predicted attack class label.

Random Forest combines multiple decision trees using majority voting:

$$\mathbf{Y} = \text{mode}(\mathbf{T}_1(\mathbf{X}), \mathbf{T}_2(\mathbf{X}), \dots, \mathbf{T}_n(\mathbf{X}))$$

RNN and LSTM models process sequential data using recurrent hidden states:

$$\mathbf{h}_t = f(\mathbf{W}\mathbf{x}_t + \mathbf{U}\mathbf{h}_{t-1} + \mathbf{b})$$

Transformer models use self-attention mechanisms represented as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\mathbf{Q}\mathbf{K}^T / \sqrt{d_k})\mathbf{V}$$

SHAP-based Explainable AI computes feature contribution scores to explain model predictions and identify important intrusion detection features.

CHAPTER - 4 - IMPLEMENTATION

The proposed AI-Powered Intrusion Detection System was implemented using machine learning, deep learning, transformer architectures, and Explainable AI techniques for multi-class cyberattack detection. The implementation process involved dataset preprocessing, feature engineering, model training, performance evaluation, and Explainable AI analysis.

4.1 Datasets Used

Two benchmark cybersecurity datasets were used in this project:

1. UNSW-NB15 Dataset

2. CIC-IDS Collection Dataset

These datasets contain both benign and malicious network traffic records across multiple cyberattack categories such as DoS, DDoS, Botnet, Brute Force, PortScan, and Web attacks. The datasets include network flow features such as packet size, flow duration, packet timing, packet rate, and traffic statistics.

4.2 Data Preprocessing

Before model training, several preprocessing techniques were applied to improve data quality and model performance.

The preprocessing steps included:

- * Removing missing and duplicate values
- * Label encoding for attack categories
- * Feature scaling using normalization techniques
- * Train-test data splitting
- * Dataset balancing for handling class imbalance

Feature scaling was especially important for deep learning and transformer-based models to improve training stability and convergence.

4.3 Machine Learning Models

Several machine learning algorithms were implemented for intrusion detection:

- **Decision Tree**

Decision Tree classifiers were used for basic attack classification using hierarchical decision rules.

- **Random Forest**

Random Forest combines multiple decision trees using ensemble learning and majority voting to improve classification accuracy and reduce overfitting.

- **XGBoost**

XGBoost uses gradient boosting techniques to optimize classification performance and improve attack detection capability.

These models were implemented using the Scikit-learn and XGBoost libraries in Python.

4.4 Deep Learning Models

Deep learning architectures were implemented using TensorFlow and Keras frameworks.

- **Recurrent Neural Network (RNN)**

RNN models were used to learn sequential patterns and temporal relationships in network traffic behaviour.

- **Long Short-Term Memory (LSTM)**

LSTM models improved long-term dependency learning using memory cells and gating mechanisms. Dropout layers were added to reduce overfitting and improve model generalization.

The deep learning models were trained using categorical cross-entropy loss functions and the Adam optimizer.

4.5 Transformer-Based Architecture

A Transformer-based intrusion detection model was implemented using Multi-Head Attention layers and dense neural network layers. The Transformer architecture used self-attention mechanisms to learn contextual relationships between network traffic features.

The implementation included:

- * Multi-Head Attention Layer
- * Layer Normalization
- * Global Average Pooling
- * Dense Layers with Dropout

Although the Transformer model demonstrated strong learning capability, its performance was comparatively lower than ensemble learning approaches on structured tabular network traffic data.

4.6 Explainable Artificial Intelligence

To improve model interpretability and transparency, SHAP (SHapley Additive exPlanations) was integrated into the intrusion detection framework.

SHAP analysis was used to:

- * Identify important network traffic features
- * Explain model predictions
- * Improve transparency of intrusion detection decisions

The SHAP feature importance plots showed that packet size, flow statistics, and packet timing-related features significantly influenced cyberattack classification.

4.7 Evaluation Metrics

The implemented models were evaluated using the following performance metrics:

- * Accuracy
- * Precision
- * Recall
- * F1-score
- * ROC-AUC
- * Confusion Matrix Analysis

These metrics helped compare the performance of machine learning, deep learning, and transformer-based intrusion detection models.

4.8 Tools and Technologies Used

The implementation was carried out using the following tools and technologies:

- * Python Programming Language
- * Google Colab
- * TensorFlow and Keras
- * Scikit-learn
- * XGBoost

- * SHAP Library
- * Pandas and NumPy
- * Matplotlib and Seaborn

The complete implementation workflow involved dataset preprocessing, model training, performance evaluation, visualization, and Explainable AI analysis for developing an intelligent and interpretable intrusion detection system

CHAPTER - 5 - RESULTS AND ANALYSIS

The proposed AI-Powered Intrusion Detection System was evaluated using the UNSW-NB15 and CIC-IDS datasets for multi-class cyberattack detection. Multiple machine learning, deep learning, and transformer-based models were trained and compared using evaluation metrics such as Accuracy, Precision, Recall, F1-score, ROC-AUC, and confusion matrix analysis.

The experimental results demonstrated that ensemble learning models such as Random Forest and XGBoost achieved the best overall intrusion detection performance. These models effectively learned complex attack patterns from structured network traffic data and produced high Accuracy, F1-score, and ROC-AUC values across multiple attack categories. Random Forest particularly showed stable and reliable classification performance with minimal misclassification errors.

Decision Tree models provided comparatively lower performance because of their tendency to overfit complex network traffic patterns. However, they still demonstrated effective attack classification capability for several intrusion categories.

Deep learning models including Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks demonstrated strong sequential learning capability for intrusion detection. RNN models effectively captured temporal relationships in network traffic behavior, while LSTM models improved long-term dependency learning using memory cells and gating mechanisms. Both models achieved high ROC-AUC scores and strong attack classification performance.

The Transformer-based intrusion detection model used Multi-Head Attention mechanisms to learn contextual relationships between network traffic features. Although the Transformer architecture

demonstrated effective learning capability, its performance remained comparatively lower than ensemble learning approaches. This occurred because the datasets mainly contained structured tabular network traffic features rather than highly sequential contextual data where Transformers typically perform best.

The final comparison graph and evaluation table illustrated the performance differences between all implemented models. Experimental analysis showed that ensemble learning approaches outperformed more complex deep learning and transformer architectures on structured intrusion detection datasets.

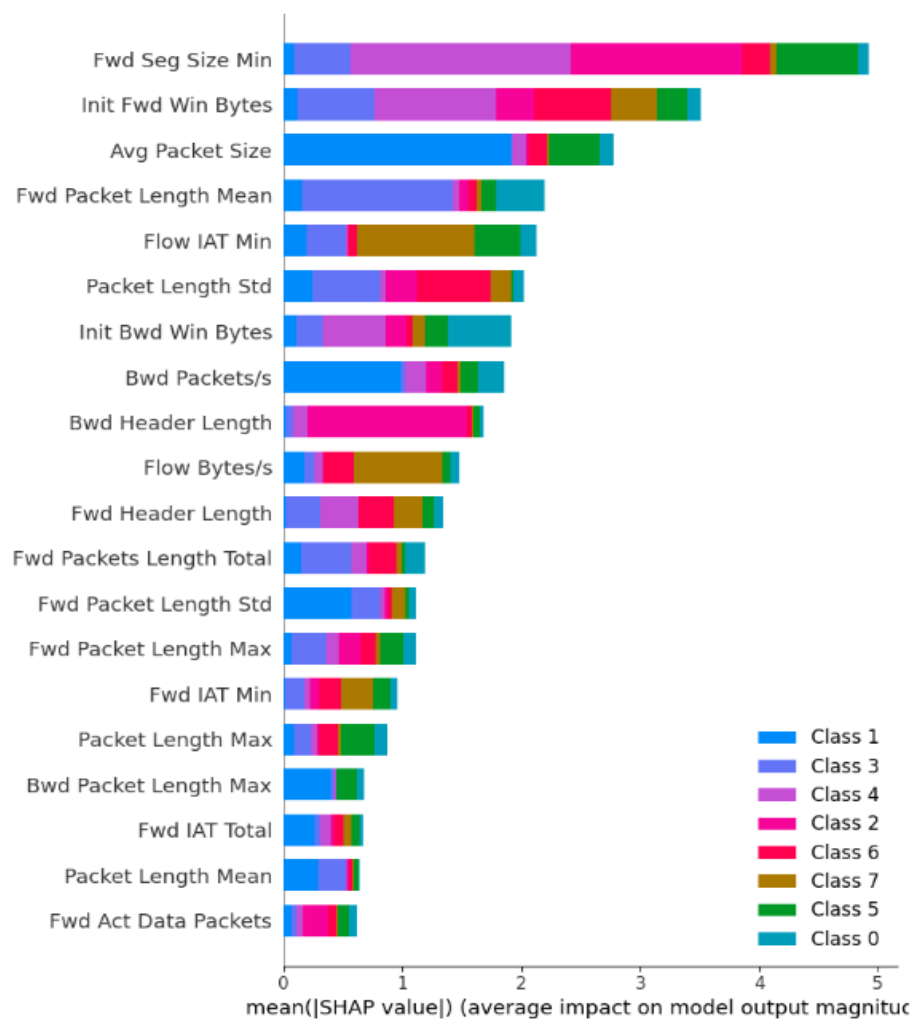
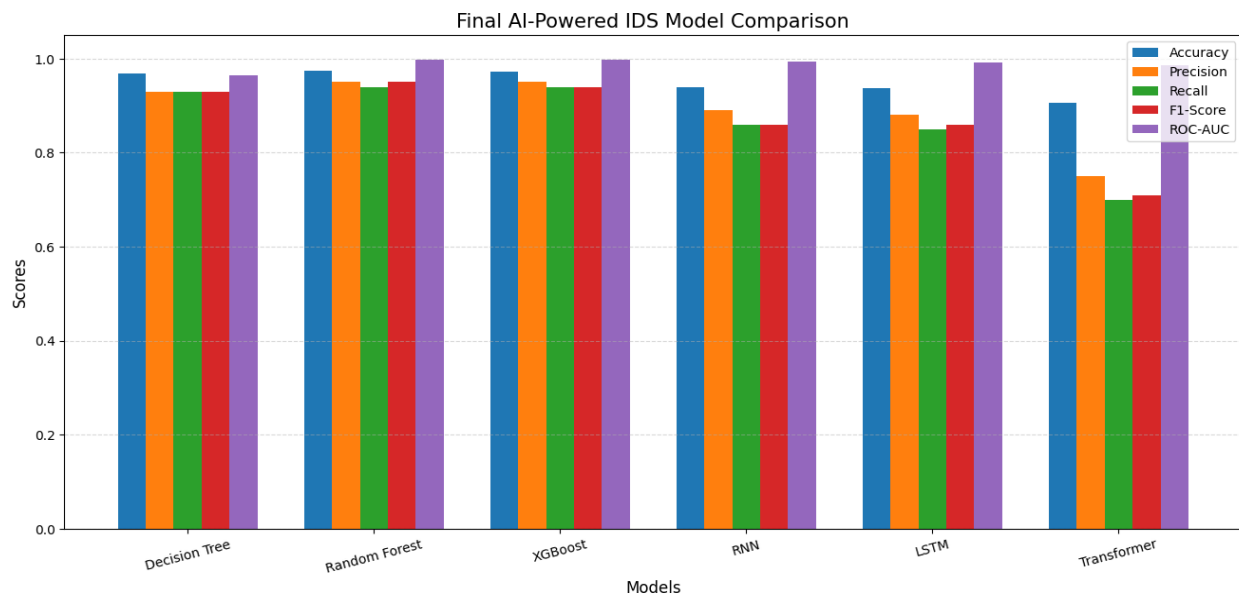
Confusion matrix analysis further demonstrated that the Random Forest model effectively classified both benign and malicious traffic with minimal false classifications. Most attack categories were accurately detected, although a few minority attack classes showed slightly lower detection performance because of class imbalance in the datasets.

SHAP-based Explainable Artificial Intelligence was integrated to improve model transparency and interpretability. SHAP feature importance analysis identified several network traffic features such as packet size, flow statistics, and packet timing as the most influential contributors to intrusion detection decisions. This analysis helped explain how the AI models differentiated between benign and malicious traffic.

Overall, the proposed AI-Powered Intrusion Detection System successfully combined machine learning, deep learning, transformer architectures, and Explainable AI techniques to achieve accurate, scalable, and interpretable cyberattack detection on benchmark cybersecurity datasets.

FINAL MODEL COMPARISON TABLE:

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	Decision Tree	0.9688	0.93	0.93	0.93	0.9649
1	Random Forest	0.9746	0.95	0.94	0.95	0.9972
2	XGBoost	0.9715	0.95	0.94	0.94	0.9978
3	RNN	0.9394	0.89	0.86	0.86	0.9930
4	LSTM	0.9366	0.88	0.85	0.86	0.9922
5	Transformer	0.9059	0.75	0.70	0.71	0.9850



CHAPTER - 6 - CONCLUSION & FUTURE WORK

This project successfully developed an AI-Powered Intrusion Detection System using machine learning, deep learning, transformer architectures, and Explainable AI techniques for multi-class cyberattack detection. The proposed system was evaluated using the UNSW-NB15 and CIC-IDS datasets containing both normal and malicious network traffic records.

Experimental results demonstrated that ensemble learning approaches such as Random Forest and XGBoost achieved the best overall intrusion detection performance with high Accuracy, F1-score, and ROC-AUC values. Deep learning models such as RNN and LSTM also demonstrated strong sequential learning capability, while the Transformer model showed comparatively lower performance on structured tabular network traffic data.

The integration of SHAP-based Explainable Artificial Intelligence improved model transparency and interpretability by identifying the most influential network traffic features contributing to intrusion detection decisions. The project demonstrated the effectiveness of AI-driven cybersecurity solutions for detecting modern cyber threats accurately and efficiently.

In future work, the proposed intrusion detection framework can be extended for real-time network monitoring and deployment in practical cybersecurity environments. Hybrid deep learning and transformer-based architectures can also be explored to further improve intrusion detection performance. Additional improvements may include federated learning, lightweight IDS models for IoT environments, and evaluation using live network traffic data for enhanced scalability and robustness.

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